



BHABHA UNIVERSITY Bhopal

Scheme of Examination

Second Semester- Master of Technology (Power Systems) Session 2018-19

SPECIALIZATION – POWER SYSTEM (PS)

S.No .	Subject Code	Subject Name	Periods per week			Credits	Maximum Marks (Theory Slot)			Maximum Marks (Practical Slot)		Total Marks
			L	T	P		End Sem. Exam.	Tests (Two)	Assign ments /Quiz	End Sem. Practical /Viva	Practical Record/ assignm ent/Quiz /Present ation	
1.	EE-201	Reactive Power Control & FACTS	3	1	-	4	70	20	10	-	-	100
2.	EE-202	Power Quality and Conditioning	3	1	-	4	70	20	10	-	-	100
3.	EE-203	Restructed Power Systems	3	1	-	4	70	20	10	-	-	100
4.	EE-204	Power System Transients	3	1	-	4	70	20	10	-	-	100
5.	EE-205	Energy Conservation and Management	3	1	-	4	70	20	10	-	-	100
6.	EE-206	Lab-III – Computer applications in Power Systems Lab	-	-	6	6	-	-	-	90	60	150
7.	EE-207	Lab-IV Advance Power System Lab	-	-	6	6	-	-	-	90	60	150
		Total	15	5	12	32	350	100	50	180	120	800

L: Lecture - T: Tutorial - P: Practical

End Semester Examination will be of three hours.



BHABHA UNIVERSITY, BHOPAL

COURSE CURRICULUM FACULTY OF ENGINEERING

Postgraduate Programme

(Courses effective from Academic Year 2018-19)

SYLLABUS OF COURSES TO BE OFFERED “Master of Technology”

Disclaimer: The postgraduate programme in Master of Technology syllabus & schemes is uploaded as given by the Faculty concerned to the Academic Council. The same has been approved as it is by the Academic Council on 02.07.2018.

Dean faculty of Engineering, Bhabha University

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Master of Technology
Branch - Electrical Engineering
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EE-201 Reactive Power Control & FACTS

UNIT-I

Introduction to FACTS, Static Var Compensator (SVC), Principle of operation, configuration and control. Thyristor Controlled Series compensator (TCSC), Interline Power Flow Controller (IPFC): Principle of operation, configuration and control, Unified Power Flow Controller (UPFC): Principle of operation, configuration and control, Simulation of UPFC, Steady state model of UPFC, Application for mitigation of SSR, Static Compensator (STATCOM), Static Synchronous Series Compensator (SSSC).

UNIT-II

Linearised model of power systems installed with FACTS based Stabilisers, Heffron-Phillips model of a SMIB system installed with SVC, TCSC and TCPS, Heffron-Phillips model of a SMIB system with UPFC, Heffron-Phillips model of a Multi-machine system installed with SVC, TCSC and TCPS.

UNIT-III

Analysis and Design of FACTS based stabilizers, Analysis of damping torque contribution by FACTS based stabilisers installed in SMIB systems, Design of robust FACTS based stabilisers installed in SMIB systems by phase compensation method, General Structure of the FACTS devices control.

UNIT-IV

Transient Stability control with FACTS, Analysis of Power systems installed with FACTS devices, Power transmission control using Controllable Series Compensation(CSC), Power Transmission Control using SSSC, Power Transmission Control using UPFC, Power Transmission Control using Phase Shifting Transformer(PST), Power Transmission Control using UPFC.

REFERENCES BOOKS:

1. Reactive Power Control in Power Systems, T J E Miller John Wiley
2. Understanding FACTS' N G Hingorani and L Gyugyi, IEEE Press.
3. Flexible ac Transmission Systems (FACTS), Y.H. Song, A.T.Johns,IEE

Master of Technology
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EE –202 Power Quality & Conditioning

UNIT-I

Understanding Power quality, types of power quality disturbances, power quality indices, Causes and effects of power quality disturbances.

UNIT-II

Radio interference, supply standards, elimination/suppression of harmonics, classical solutions & their drawbacks, passive input filters, design of harmonic filters, Improved power quality converter topologies, (single and three phase), transformer connections

UNIT-III

Causes and effects of harmonics, converter configuration and their contribution to supply harmonics, other sources of harmonics.

UNIT-IV

Active wave shaping of input line current, constant frequency control, constant tolerance band control, variable tolerance band control, discontinuous current control, Electromagnetic interference (EMI), EMI generation, EMI standards, and elimination.

REFERENCE BOOKS:

1. Power Quality– by R.C. Duggan.
2. Power electronic converter harmonics – by Derek A. Paice.

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EE – 203 Restructured Power Systems

UNIT-I

Regulation and Deregulation, condition for deregulation, problems with regulation, risk management, congestion management, ATC, screening curve.

UNIT-II

What is competition, efficiency of perfect competition, marginal cost in power market, role of marginal cost, working with marginal cost, results of marginal cost.

UNIT-III

Define market power, price quality outcomes, three stages of market power, using price quality outcomes to show power, monopoly in power auction, market power on demand side.

UNIT-IV

Fundamentals of restructured system, Market Architecture, Load Elasticity, Social welfare maximization, Optimal Bidding, Transmission Pricing and Tracing of power, Ancillary Services, Standard Market Design.

UNIT-V

Distributed Generation in restructured markets, Developments in India, IT applications in restructured markets, Working of restructured power systems PJM.

REFERENCE BOOKS:-

1. Power system economics-designing for electricity-steven stoft. (IEEE press & WILEYINTERSCIENCE)
2. Restructured electrical power systems: operation, trading and volatility
Mohammad Shahidehpour, Muwaffaq Alomoush, Marcel Dekker Pub.,
2001

Master of Technology
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EE-204 Power System Transients

UNIT-I

Origin and nature of transients and surges. Equivalent circuit representations. Lumped and distributed circuit transients. Line energisation and de-energisation transients. Earth and earthwire effects.

UNIT-II

Insulation Coordination overvoltage limiting devices, dielectric properties, breakdown of gaseous insulation, tracking and erosion of insulation, high current arcs.

UNIT-III

Current chopping in circuit breakers. Short line fault condition and its relation to circuit breaker duty. Trapped charge effects. Effect of source and source representation in short line fault studies. Control of transients.

UNIT-IV

Lightning phenomena, Influence of tower footing resistance and earth resistance. Traveling waves in distributed parameter multi-conductor lines, parameters as a function of frequency.

UNIT-V

Simulation of surge diverters in transient analysis. Influence of pole opening and pole closing. Fourier integral and Z transform methods in power system transients.

REFERENCE BOOKS :-

1. Power System Transients by C. S. Indulkar and D.P. Kothar
2. Power Stemys Transients by Vanikov

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EE-205 Energy Conservation & Management

UNIT-I

Energy audit, Energy monitoring, Energy accounting and analysis, Auditing and targeting. Energy conservation policy, Energy management & audit, Energy audit, Types of energy audit, energy management (audit), qualities and function of energy managers, language of an energy manager, Questionnaire, Check list for top management, Loss of energy in material flow, energy performance, Material load energy balance diagram.

UNIT-II

Energy conservation task before industry, Energy conservation equipments, Co Generation, Energy conservation in Sugar, Textiles, Cement, process industry, Electrical Energy Conservation in building, heating, lighting, domestic gadgets.

UNIT-III

Energy efficient electric drives, Energy efficient motors V.S.D. power factor improvement in power system, Energy Conservation in transportation system especially in electric vehicle. Energy flow networks, Simulation & modeling, formulation & Objective & constraints, alternative option, Matrix chart.

UNIT-IV

Thermodynamics of Energy Conservation, Basic principle, Irreversibility and second law, efficiency analysis of systems, Primary energy sources, optimum use of prime-movers, energy efficient house keeping, energy recovery in thermal systems, waste heat recovery techniques, thermal insulation, Thermal energy audit in heating, ventilation and air conditioning. Maintenance and Energy audit, friction, lubrication and tribo-logical innovations. Predictive and preventive maintenance.

UNIT-V

Load curve analysis & load management DSM, Energy storage for power systems (Mechanical, Thermal, Electrical & Magnetic) Restructuring of electric tariff from energy conservation consideration, Economic analysis depreciation method, time value of money, Evaluation method of projects, replacement analysis, special problems inflation risk analysis. Pay back period, Energy economics, Cost Benefit Risk analysis, Pay back period..

REFERENCE BOOKS:-

1. Energy management-Satya Publication-S.K.Soni.
2. Energy Management Head Book- W.C. Turner, John Wiley
3. Elec, Energy Utilization & Conservation. Dr. Tripathi S.C.
4. Energy Management – W.R. Murphy & G. Mckey Butler worths

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LIST OF EXPERIMENTS

EE-206 (Lab-III – Computer applications in Power Systems Lab)

1. Economic load dispatch neglecting losses and generator limits.
2. Load flow solution by fast decoupled method.
3. Load flow solution by GS-method.
4. Load flow solution by newton-raphson method.
5. Formation of Y-bus.
6. Z-bus building algorithm.

EE-207 (Lab-IV Advance Power System Lab)

1. Study of the following relays for Generator protection.
 - (i) Earth fault protection.
 - (ii) Differential relay.
 - (iii) Negative Sequence relay.
2. To simulate a Single Phase Thyristor Controlled Reactor using Simulink.
3. To simulate a Single Phase Thyristor Switched Capacitor using Simulink